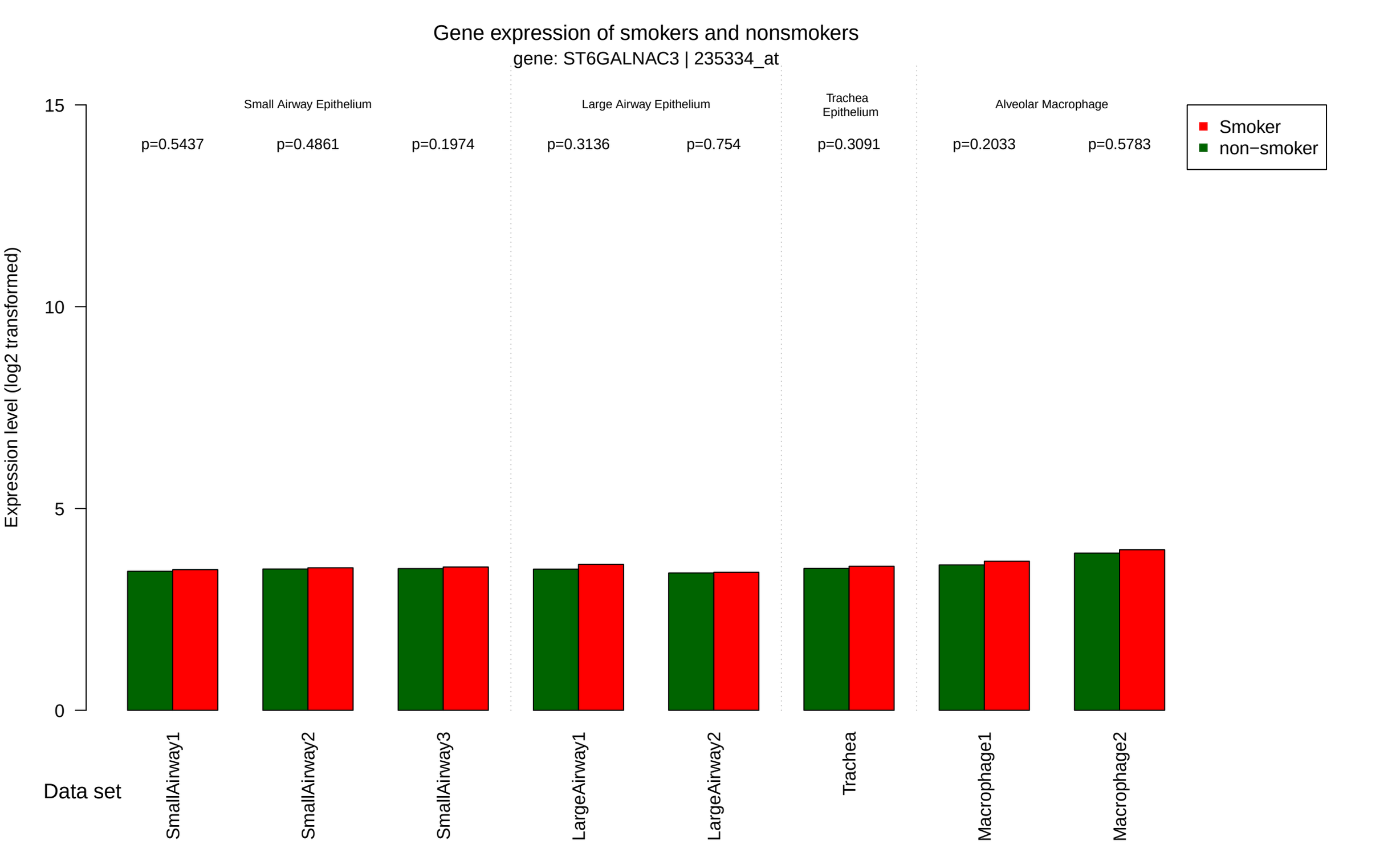
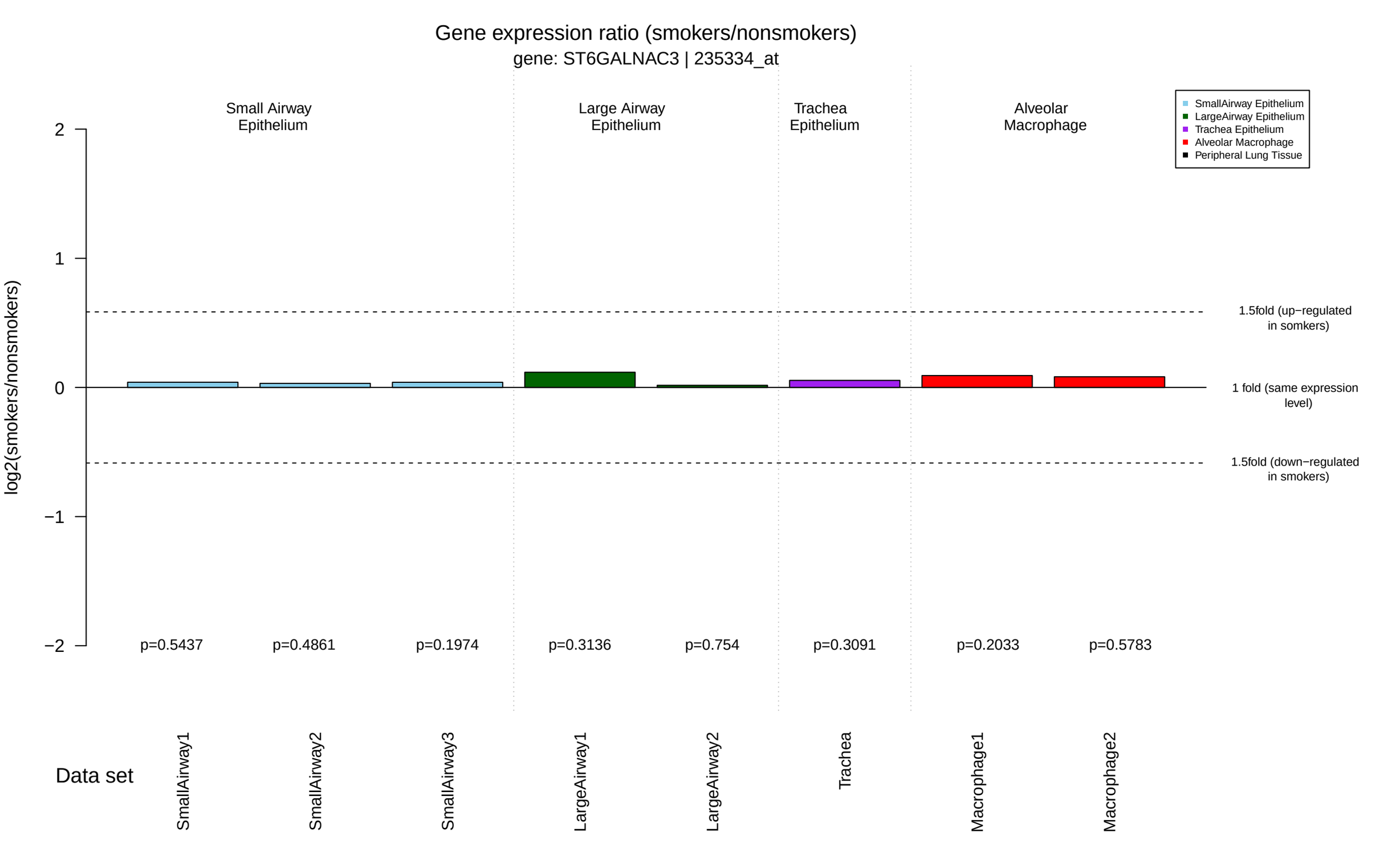
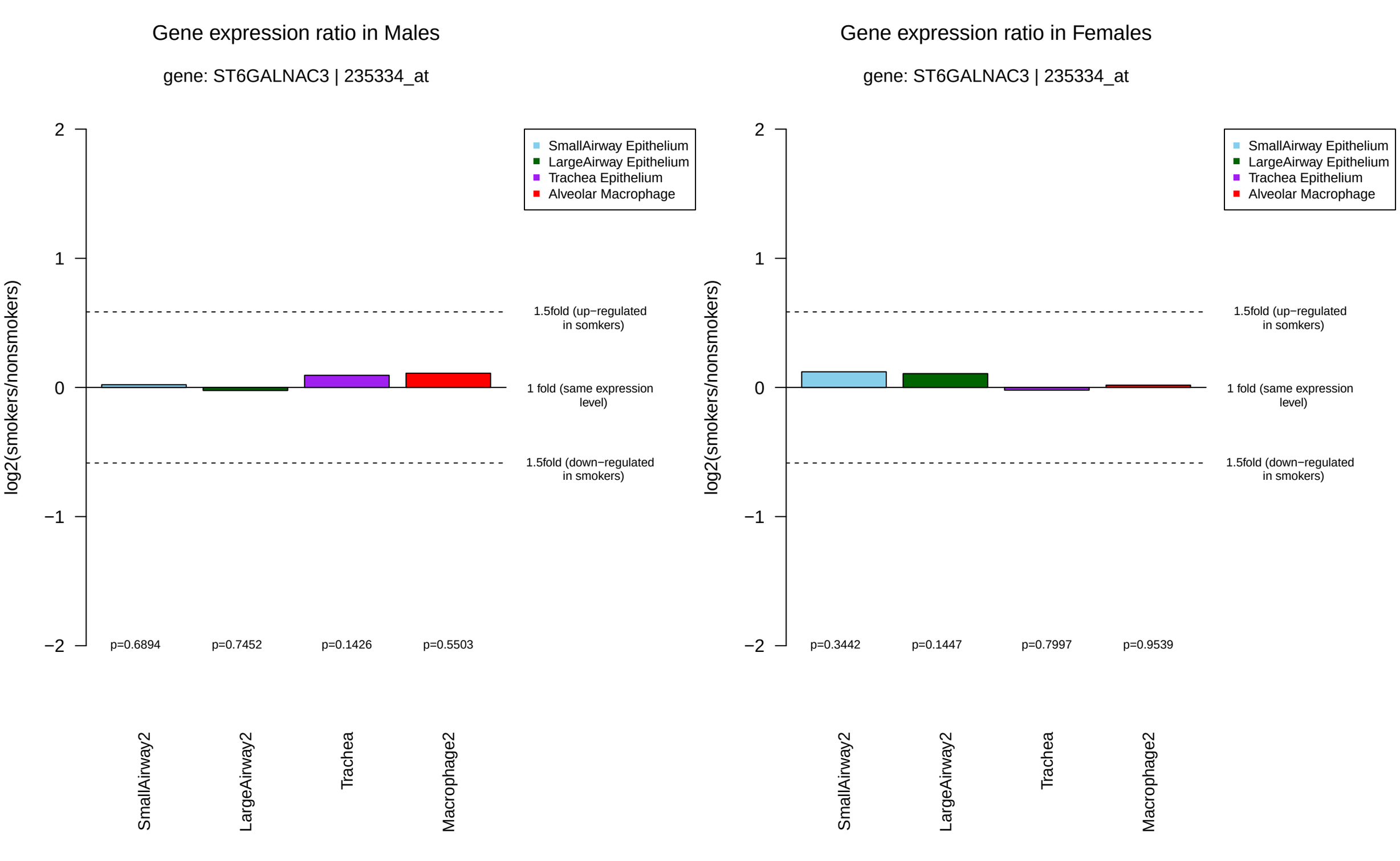
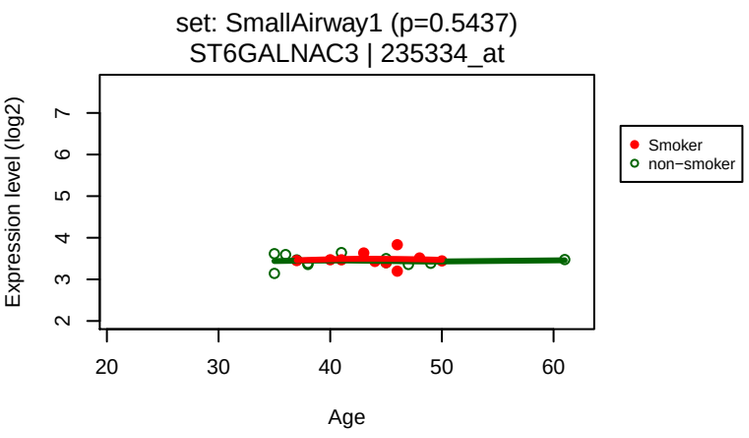
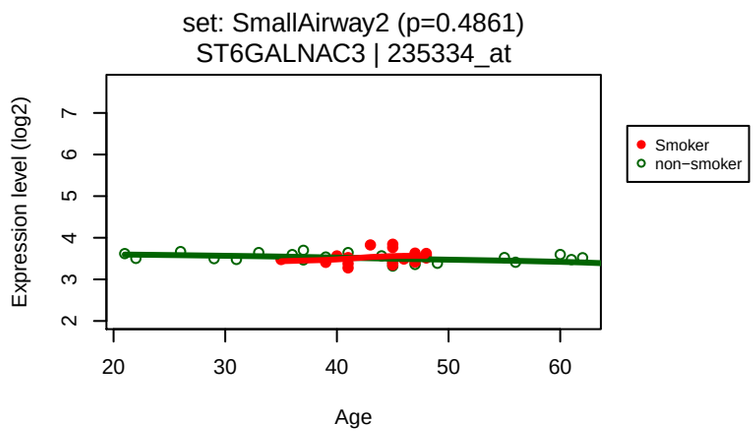
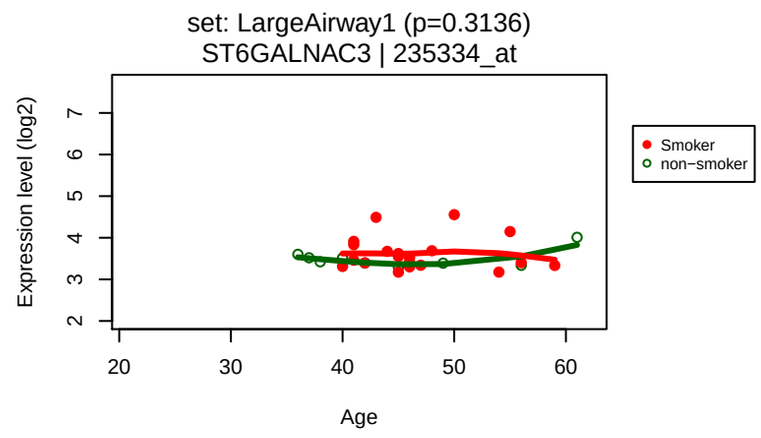

Gene expression of smokers and nonsmokers
gene: ST6GALNAC3 | 235334_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5437
p=0.4861
p=0.1974
p=0.3136
p=0.754
p=0.3091
p=0.2033
p=0.5783
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

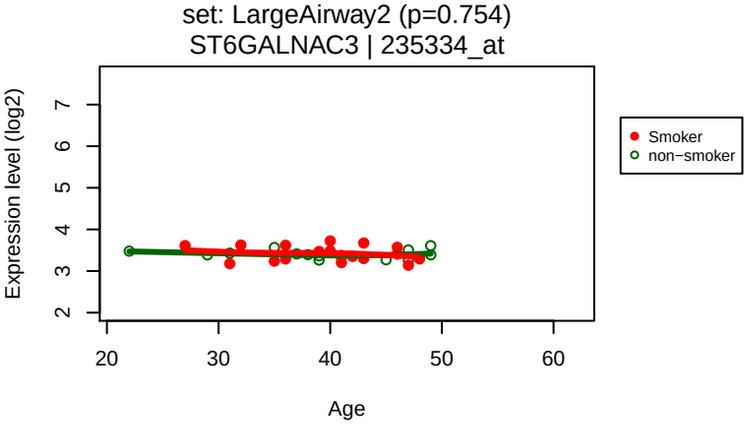

Gene expression ratio (smokers/nonsmokers)
gene: ST6GALNAC3 | 235334_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5437
p=0.4861
p=0.1974
p=0.3136
p=0.754
p=0.3091
p=0.2033
p=0.5783
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

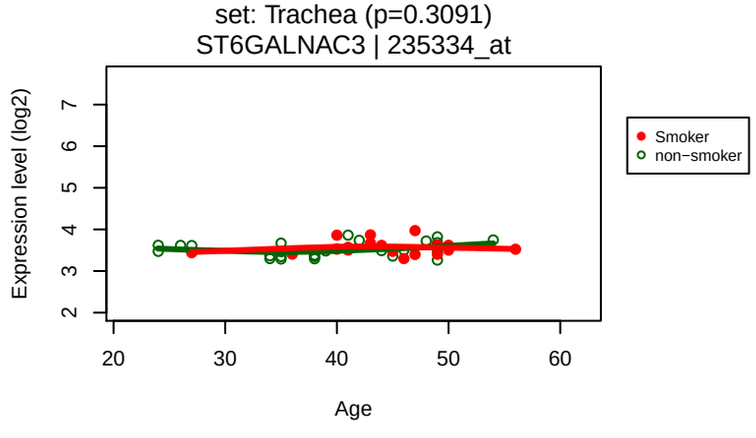

Gene expression ratio in Males
gene: ST6GALNAC3 | 235334_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6894
p=0.7452
p=0.1426
p=0.5503
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GALNAC3 | 235334_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3442
p=0.1447
p=0.7997
p=0.9539
SmallAirway2
LargeAirway2
Trachea
Macrophage2

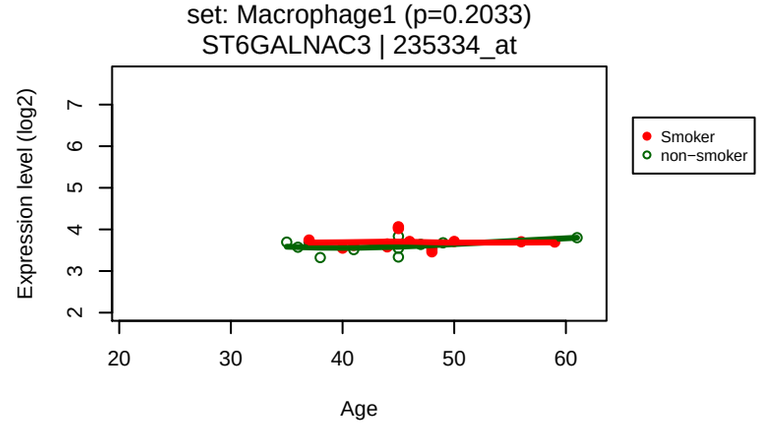

set: SmallAirway1 (p=0.5437)
ST6GALNAC3 | 235334_at
Expression level (log2)
Age
Smoker
non−smoker

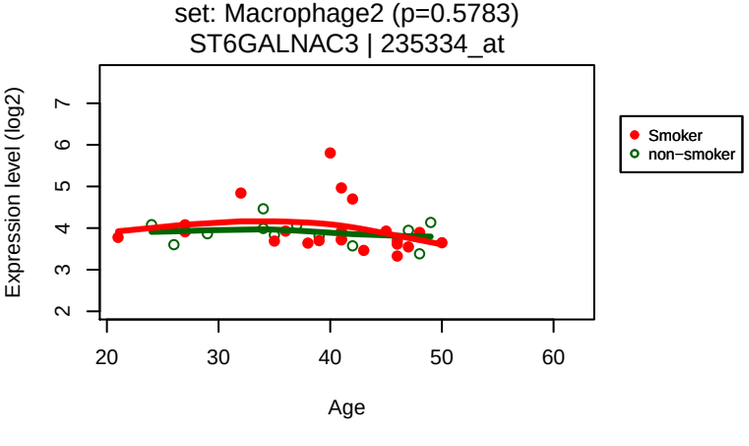

set: SmallAirway2 (p=0.4861)
ST6GALNAC3 | 235334_at
Expression level (log2)
Age
Smoker
non−smoker

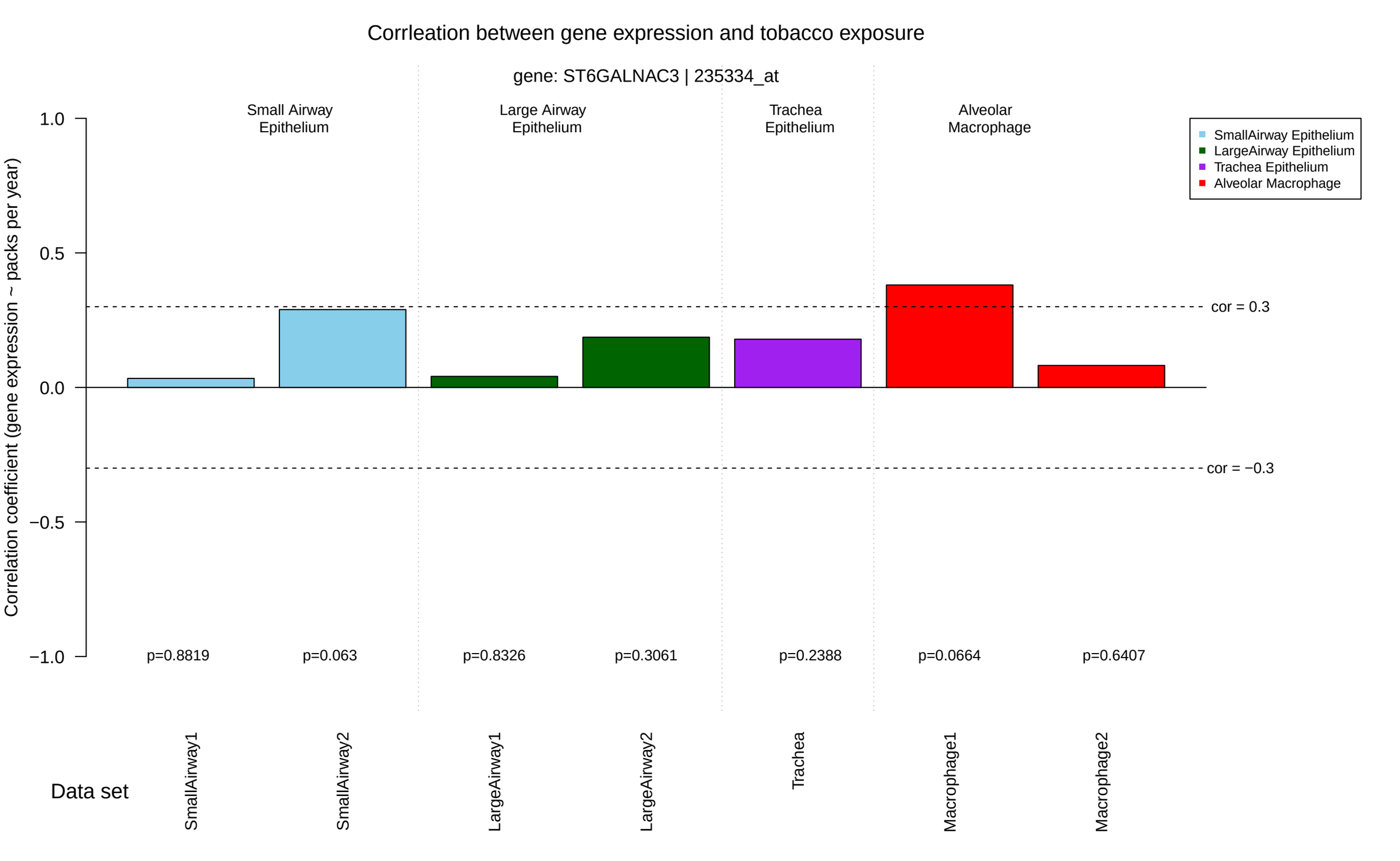

set: LargeAirway1 (p=0.3136)
ST6GALNAC3 | 235334_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.754)
ST6GALNAC3 | 235334_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3091)
ST6GALNAC3 | 235334_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2033)
ST6GALNAC3 | 235334_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5783)
ST6GALNAC3 | 235334_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GALNAC3 | 235334_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8819
p=0.063
p=0.8326
p=0.3061
p=0.2388
p=0.0664
p=0.6407
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2